PAKISTAN ADVANCED STRATEGIC & SCIENTIFIC REGISTER

A highly specialized dossier focusing exclusively on defense tech, strategic aerospace, mountain warfare choke points, and nuclear milestones.

1. Ballistic & Cruise Missile Systems (Strategic Arsenal)

System Class	Platform Name	Range / Payload Capacity	Strategic / Technical Operational Profile
Short-Range Ballistic (SRBM)	**Ghaznavi (Hatf-III)**	290 km 700 kg	Solid-fuel, hyper-mobile tactical system optimized for rapid deployment and strike capabilities.
Medium-Range Ballistic	**Shaheen-II (Hatf-VI)**	2,000 km 1,050 kg	Two-stage solid-propellant road-mobile ballistic system, serving as a core pillar of strategic deterrence.
Medium-Range Ballistic	**Ababeel**	2,200 km Multiple	Equipped with Multiple Independently Targetable Re-entry Vehicles (MIRV) technology to defeat missile defense nets.
Subsonic Cruise Missile	**Babur-3 (SLCM)**	450 km 450 kg	Submarine-launched cruise variant deployed from Agosta-90B submarines, completing the maritime second-strike triad.

2. Strategic Choke Points & High-Altitude Mountain Passes

Pakistan's northern and western frontiers contain some of the highest and most strategically sensitive mountain corridors in global military geometry.

Mountain Pass	Elevation	Geographic Connectivity	Strategic Military Significance
Khunjerab Pass	4,693 m	Karakoram Range; Links Gilgit-Baltistan to Xinjiang, China	Highest paved international border crossing; central arterial gate for the China-Pakistan Economic Corridor (CPEC).
Khyber Pass	1,070 m	Spin Ghar Mountains; Links Peshawar to Kabul, Afghanistan	Historical gateway for continental invasions; vital logistics bottleneck for trade and cross-border security.
Bolan Pass	1,793 m	Toba Kakar Range; Links Quetta to Jacobabad/Sibi	Connects the Balochistan plateau to the Indus Valley plains; critical defensive rail/road bottleneck.
Lowari Pass	3,118 m	Hindu Kush; Links Dir to Chitral	Historically snow-blocked 6 months a year; now bypassed by the 10.4 km Lowari Tunnel to secure year-round border logistics.

3. Nuclear Infrastructure & Scientific Projects

Facility / Installation	Location	Primary Scientific / Strategic Mandate
Kahuta Research Labs (KRL)	Kahuta, Punjab	Gas centrifuge uranium enrichment facility; pioneered national uranium-based deterrent programs.
Khushab Nuclear Complex	Khushab, Punjab	Heavy-water plutonium production site featuring multiple operational reactors for advanced tactical yields.
Chashma Nuclear Power Plant (CHASNUPP)	Chashma, Punjab	Civil nuclear power complex operating heavily monitored PWR reactors built in collaboration with CNNC.
Chagai-I & Chagai-II Sites	Ras Koh Hills & Kharan, Balochistan	Underground nuclear test sites where five simultaneous detonations (May 28, 1998) and a subsequent test (May 30) occurred.